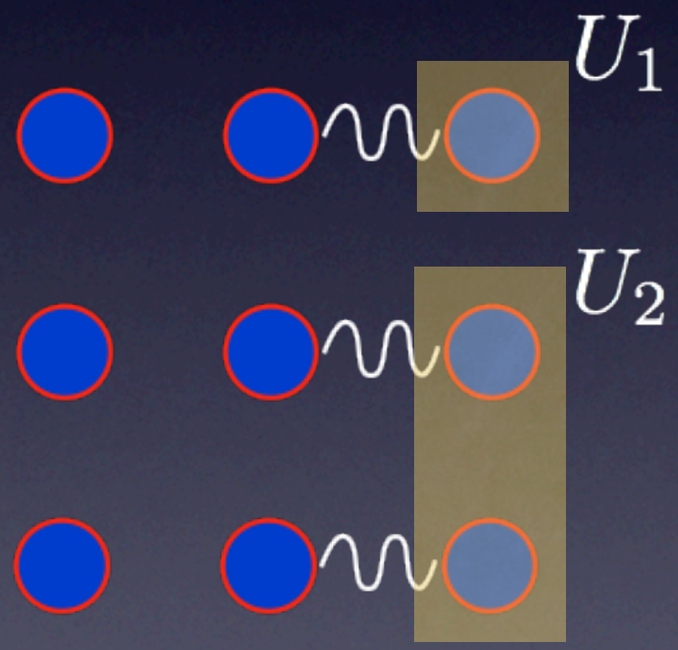
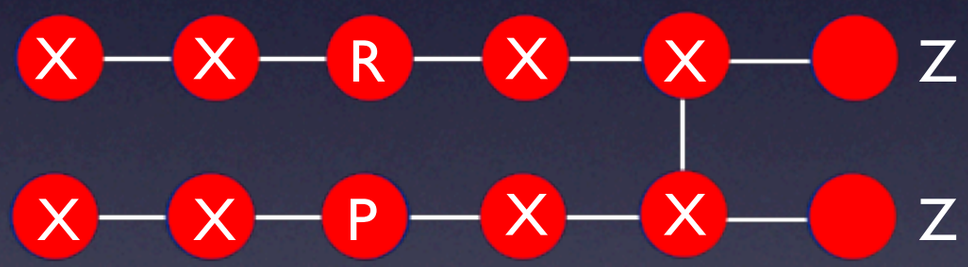
Adiabatic Cluster State Quantum Computing

- or -

Anything you can do, I can do adiabatically
I can do anything sssllllooooowwwweeeerrrrr than you





Dave Bacon
University of Washington
(in collaboration with Steve Flammia, Perimeter Institute)

Quantum Computers

1994

1995

2002

2005

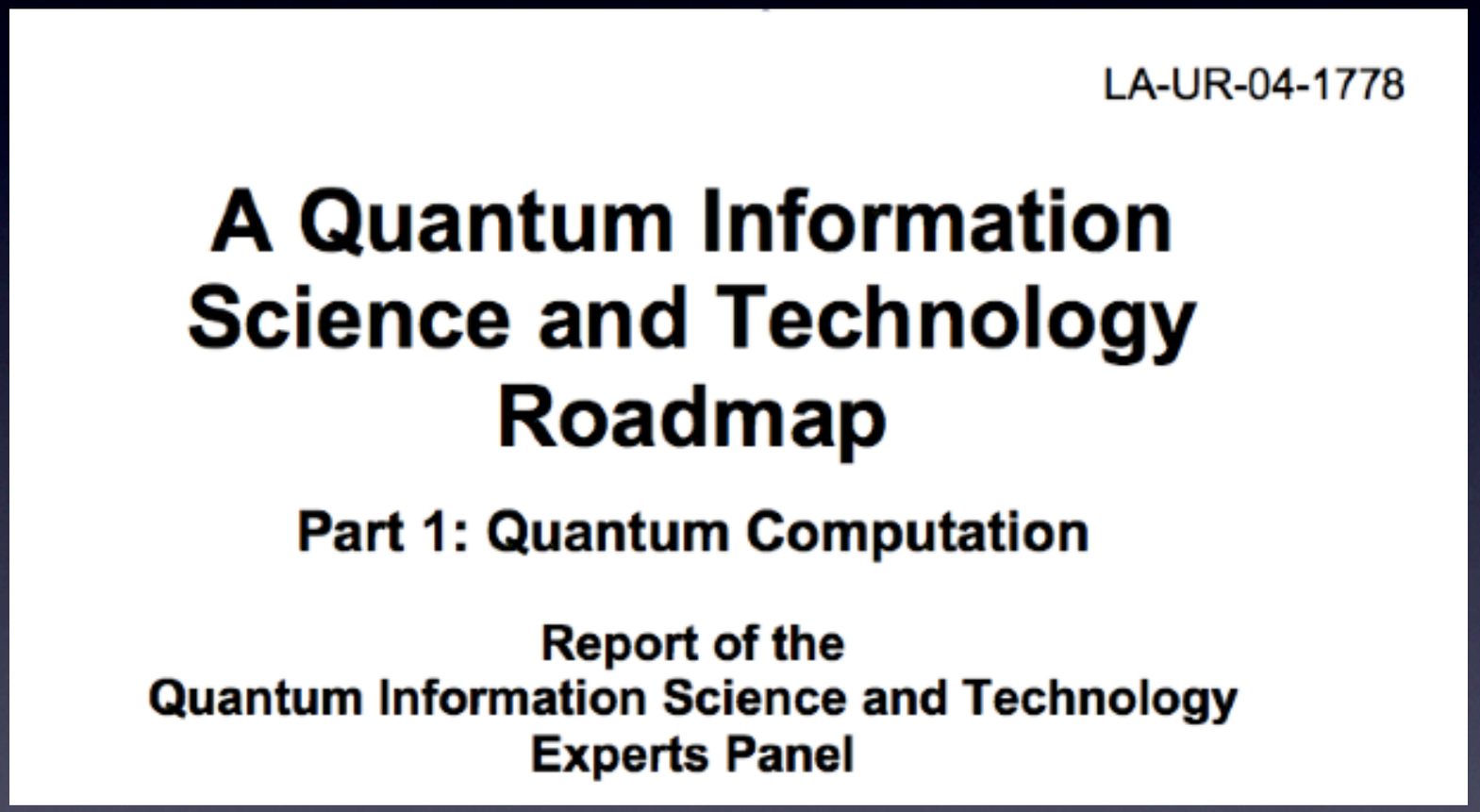
2006

Peter Shor shows that a quantum computer could

Why Don't We Have a Quantum Computer Yet?

A Plan!

2002 and 2004 roadmap:



LA-UR-04-1778

A Quantum Information Science and Technology Roadmap

Part 1: Quantum Computation

Report of the
Quantum Information Science and Technology
Experts Panel

http://qist.lanl.gov/qcomp_map.shtml

Table 4.0-1

The Mid-Level Quantum Computation Roadmap: Promise Criteria

QC Approach	The DiVincenzo Criteria: Quantum Computation #1	#2	#3	#4	#5	QC Networkability #6	#7
NMR							
Trapped Ion							
Neutral Atom							
Cavity QED							
Optical							
Solid State							
Superconducting							
Unique Qubits	This field is so diverse that it is not feasible to label the criteria with "Promise" symbols.						

1. "scalable" system of qubits
2. qubit initialization
3. long decoherence times
4. universal set of quantum gates
5. qubit measurement

achieved
proposed
no viable method known

The Roadmap



Where experimental QC dedicates most of its resources

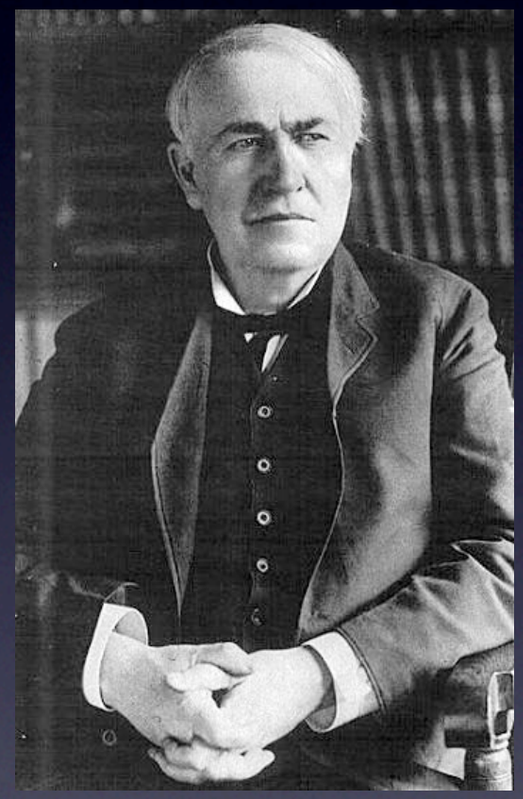
Threshold theorem

~ Given quantum computer components which fail with probability less than a threshold, one can combine these to make a robust quantum computer

Roadmaps are great for getting funding, but a horrible way to develop a technology

Why Don't We Have a Quantum Computer Yet?

"To invent, you need a good imagination, and a pile of junk"



Thomas Edison

Give Me A Junkyard ;)

Where are the teams trying to build hundreds, thousands, millions, billions of “bad qubits”?

D:wave

The Quantum Device Company™

Imagination

"To invent, you need a good imagination, and a pile of junk"
- Thomas Edison

Why should we trust that our imagination can make use of a pile of junk?

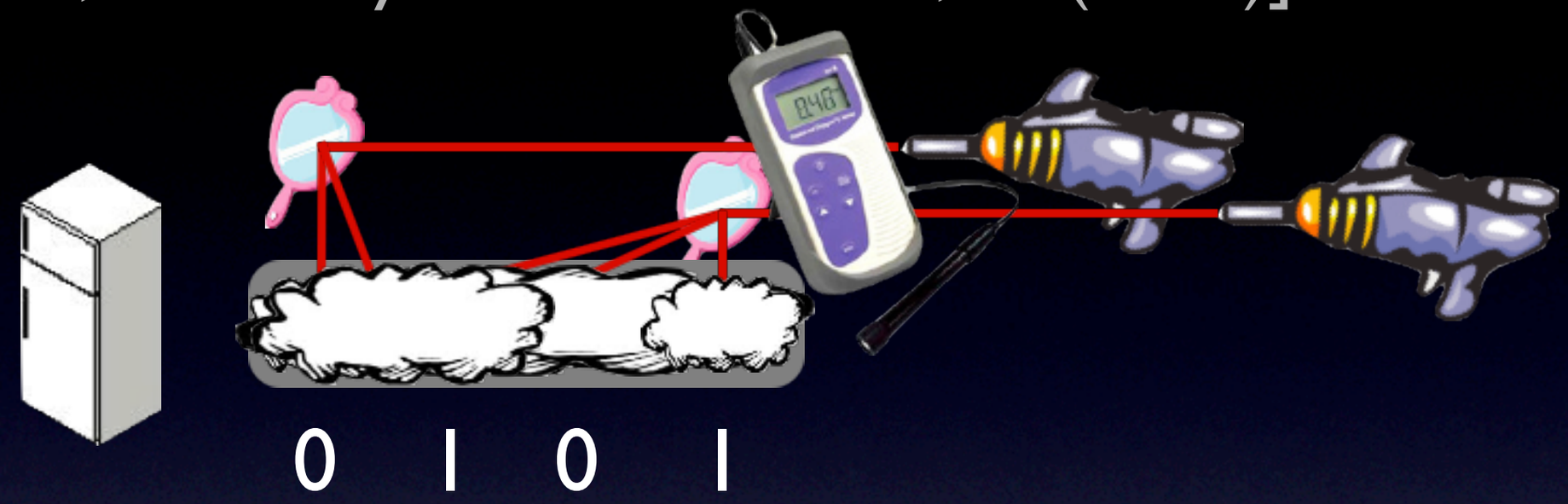
Models of Quantum Computing

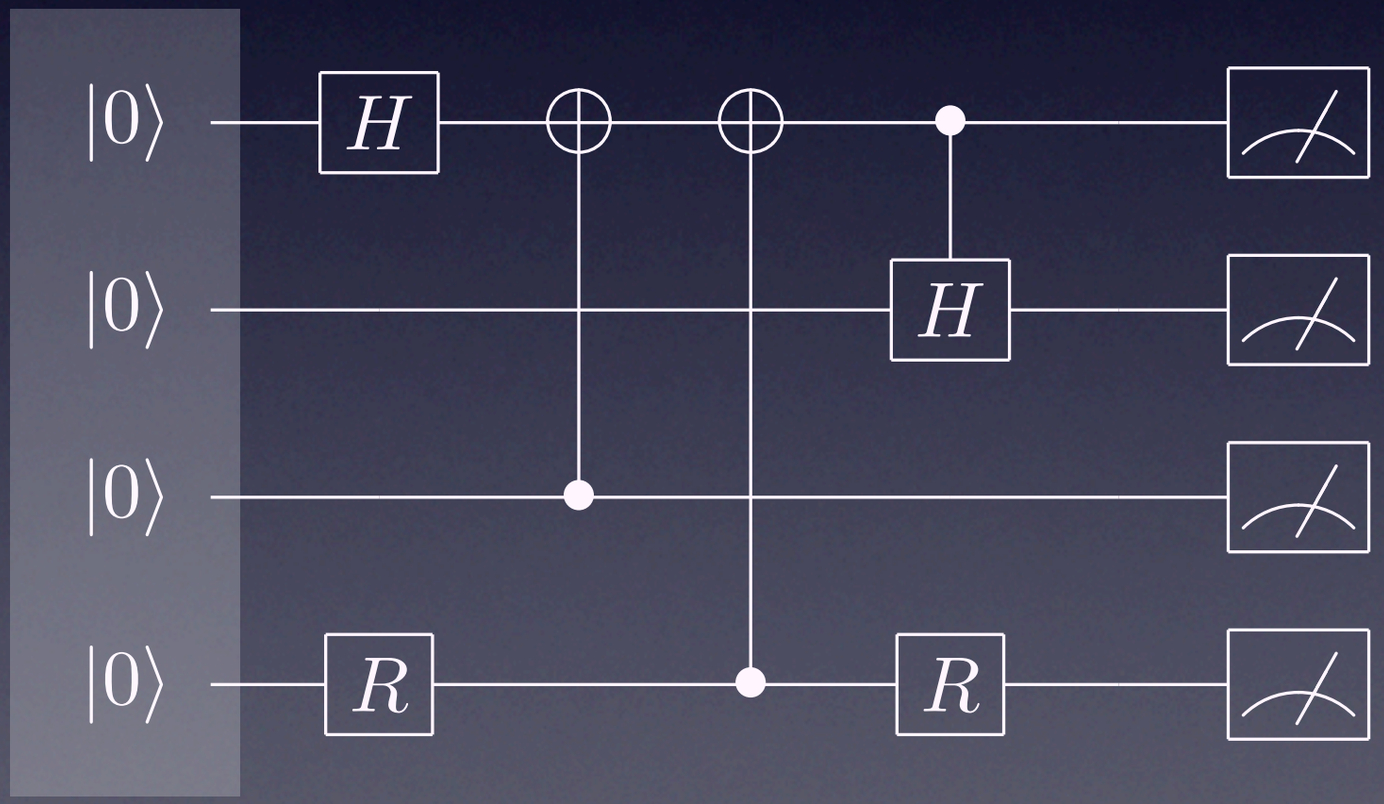
Models of QC

- Quantum Circuit Model
- Measurement Based Quantum Computing
  - One-way model
  - Teleportation model
- Universal Adiabatic Quantum Computing
- Topological Quantum Computing
- Holonomic Quantum Computing
- Quantum Random Walks

Quantum Circuit Model

[Deutsch, Proc. Roy. Soc. Lon. A 425, 73 (1989)]





Prepare states Apply gates Measure

Quantum Circuit Model

\+ Foundation of all other models

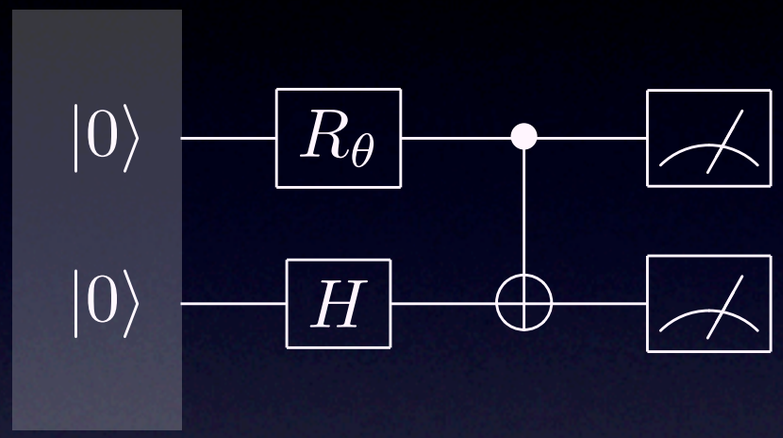
\+ Fault-tolerant

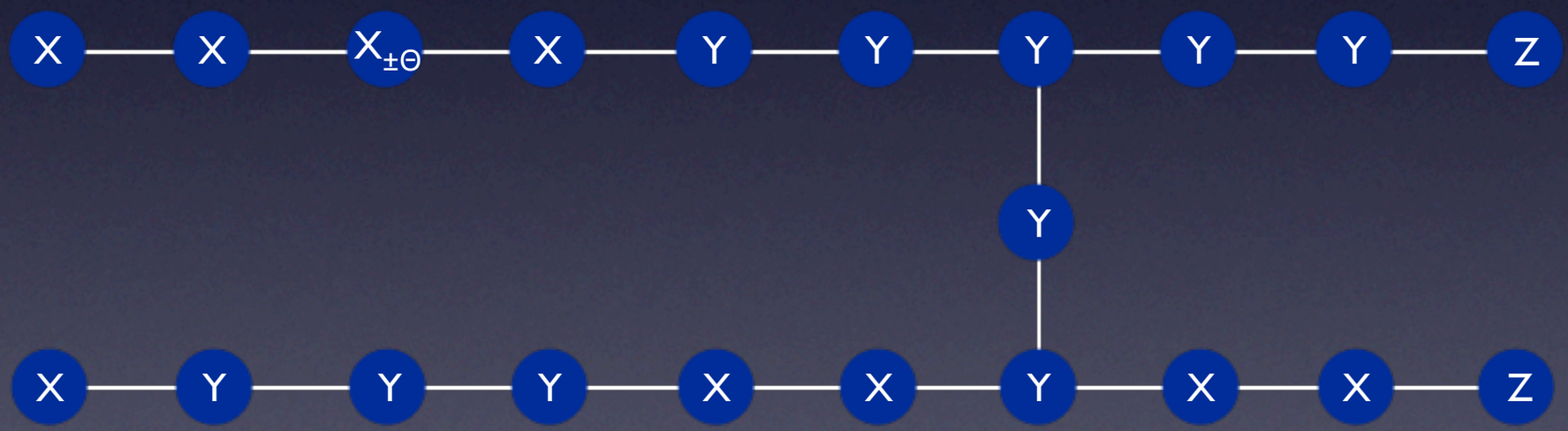
– Unitary gates require precise timing and control

– Separation into three/four procedures each requiring separate hardware

One Way QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]





Adaptive measurement

Cluster State

Entangled State

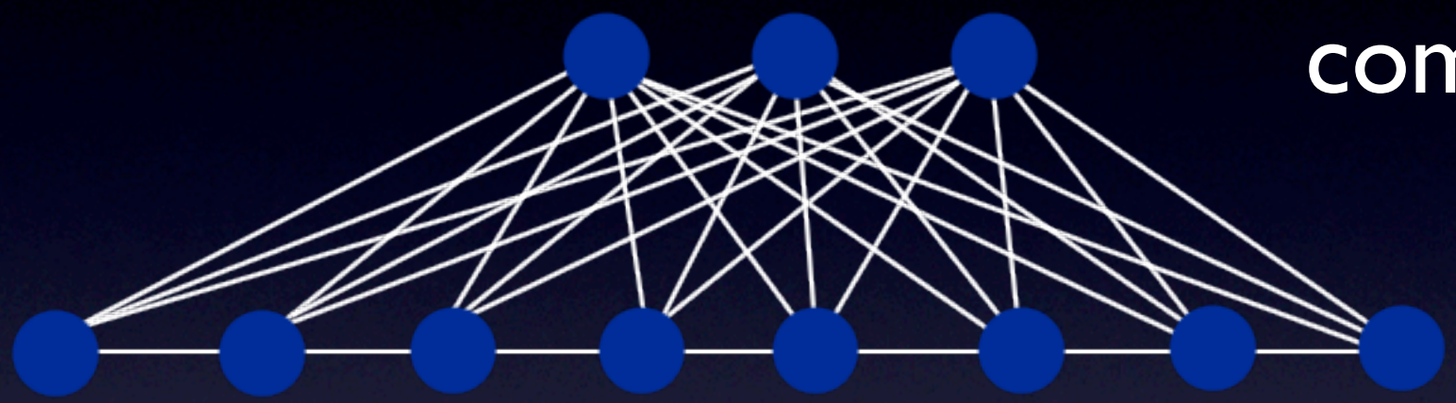
circuit

One Way QC

+ No unitary gates
+ Entanglement as “fuel” for quantum algorithm
+ Some compression in certain circuits
+ Fault-tolerant
- Measurements are often slow

Universal Adiabatic QC

[Aharonov et al. SIAM Journ. of Comp., 37 166 (2007)]



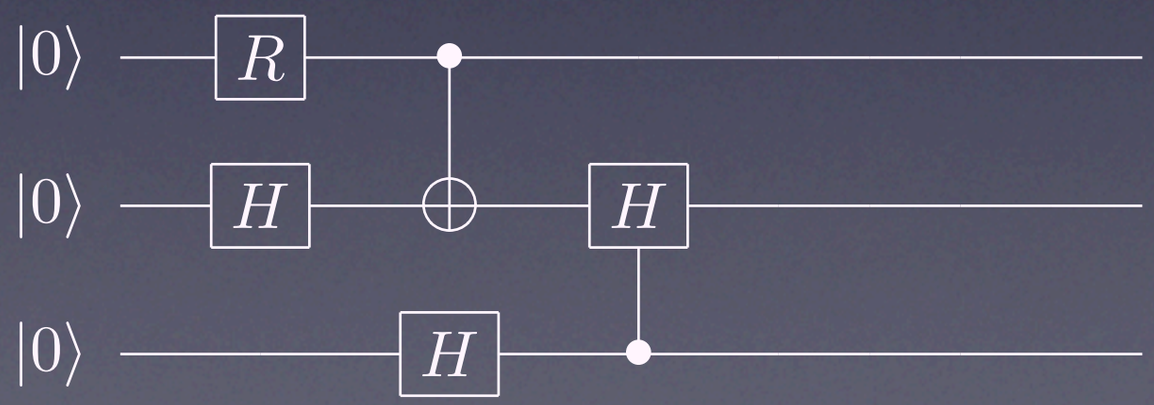
computational qubits

clock qubits

Adiabatically interpolate from H_i to H_f

$$H(t) = \left(1 - \frac{t}{T}\right) H_i + \frac{t}{T} H_f$$

Prepare system into ground state



T polynomial in depth of circuit

Final ground state:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{T} |t\rangle_c V_t |0\rangle$$

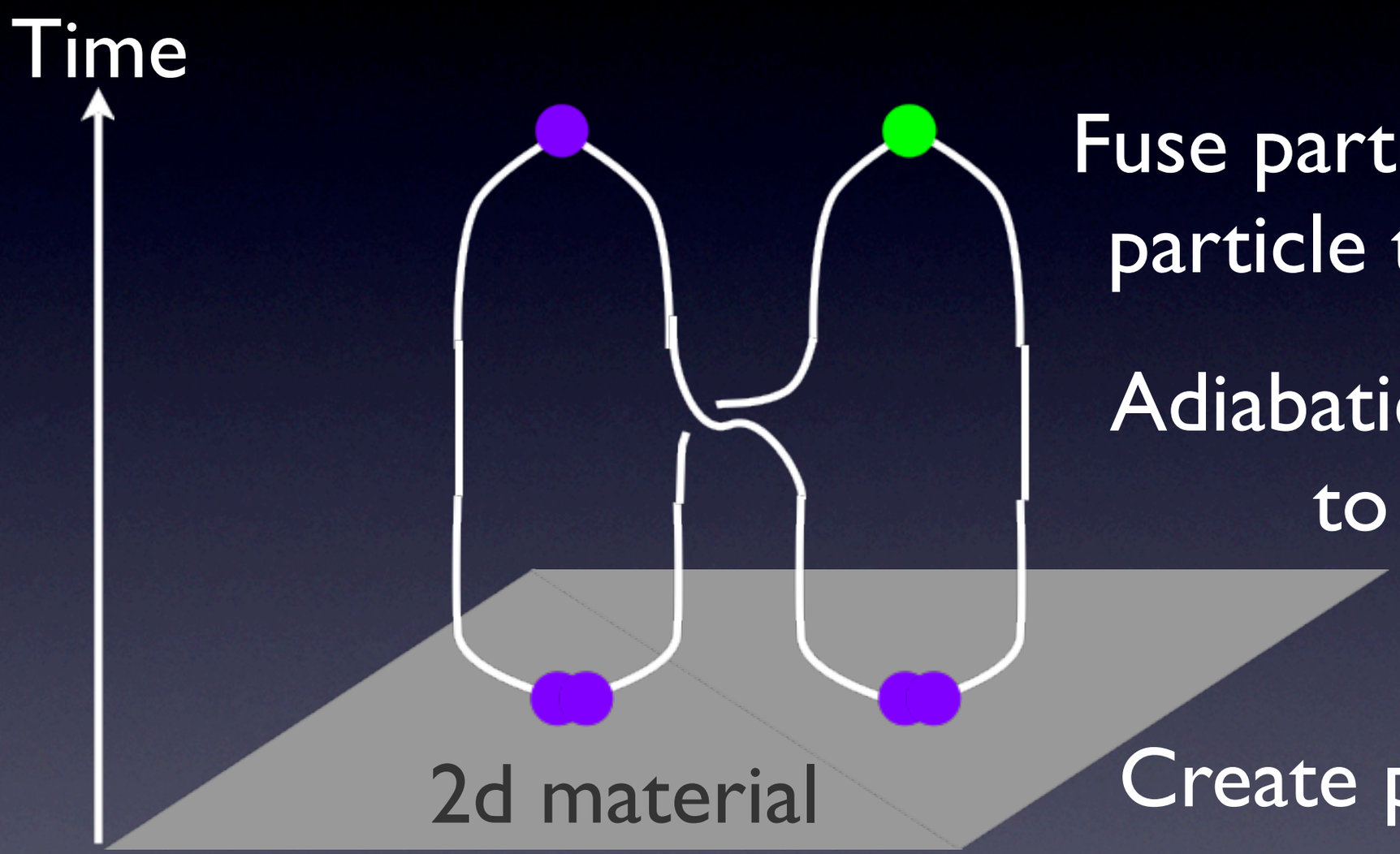
$$V_t = \prod_{i=t}^{1} U_i$$

Universal Adiabatic QC

+ Adiabatic gates - no timing issues
- Not known to be fault-tolerant
  - Current constructions use “perturbation theory gadgets” implying small energy gaps even for small circuits
  - Uber-complex

Topological QC

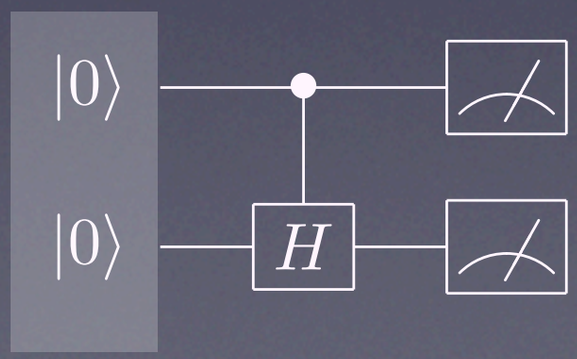
[Kitaev, Annals Phys. 303, 2 (2003), preprint (1997)]



Fuse particles and observe particle type to measure

Adiabatically braid anyons to enact gates

Create particles (anyons) out of ground state



Topological QC

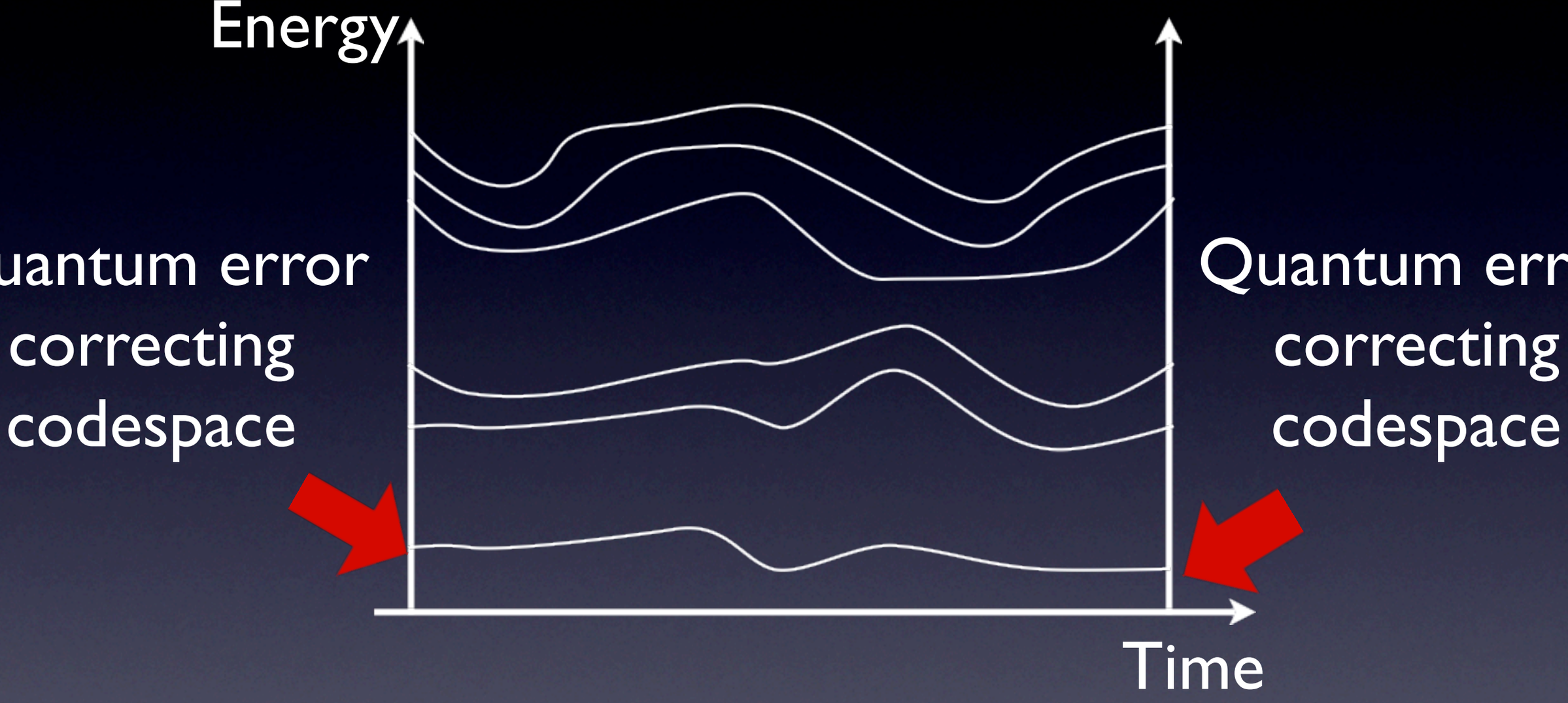
+ Adiabatic gates with topological protection
+ Low decoherence due to energy gap
+ Fault-tolerant?
- No substance (conclusively) known to support non-abelian anyons
- Fault-tolerant?

Today

Two models that combine/reinterpret many of the above models:

- Adiabatic Gate Teleportation
- Adiabatic Cluster State Quantum Computing

Adiabatic Dragging



Require: degenerate through entire evolution

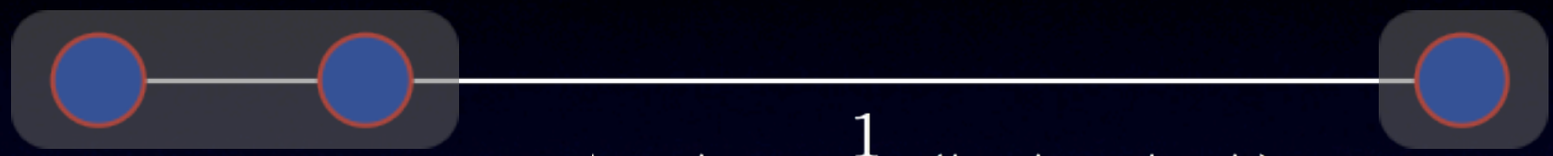
"Open loop" holonomic evolution

[D. Kult, J. Aberg, E. Sjoqvist, Phys. Rev. A 74, 022106 (2006)]

Teleportation

Teleportation Protocol

Alice



Bob

$|\Phi_{\pm}\rangle$ or $|\Psi_{\pm}\rangle$

$|\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$|\psi\rangle$

Measure qubits in Bell basis

Send 2 bits

Conditional on bits apply fixing gate

Initial Code



$|\psi\rangle \otimes |\Phi_+\rangle$

Final Code



$|\eta\rangle \otimes |\psi\rangle$

Adiabatic Teleportation

Initial Code



$|\psi\rangle \otimes |\Phi_+\rangle$

Final Code



$|\Phi_+\rangle \otimes |\psi'\rangle$

Initial Hamiltonian $\Delta > 0$ Final Hamiltonian

$H_i = -\Delta(X_2X_3 + Z_2Z_3)$ $H_f = -\Delta(X_1X_2 + Z_1Z_2)$

Evolution: $H(t) = \left(1 - \frac{t}{T}\right) H_i + \frac{t}{T} H_f$

What happens to quantum information in fist qubit?

[XX+ZZ could be XZ+ZX or XX+YY etc]

Adiabatic Teleportation

$$\begin{aligned} \bar{X}_1 &= X_1 X_2 & \bar{Z}_1 &= Z_2 Z_3 \\ \bar{X}_2 &= X_2 X_3 & \bar{Z}_2 &= Z_1 Z_2 \\ \bar{X}_3 &= X_1 X_2 X_3 & \bar{Z}_3 &= Z_1 Z_2 Z_3 \end{aligned}$$

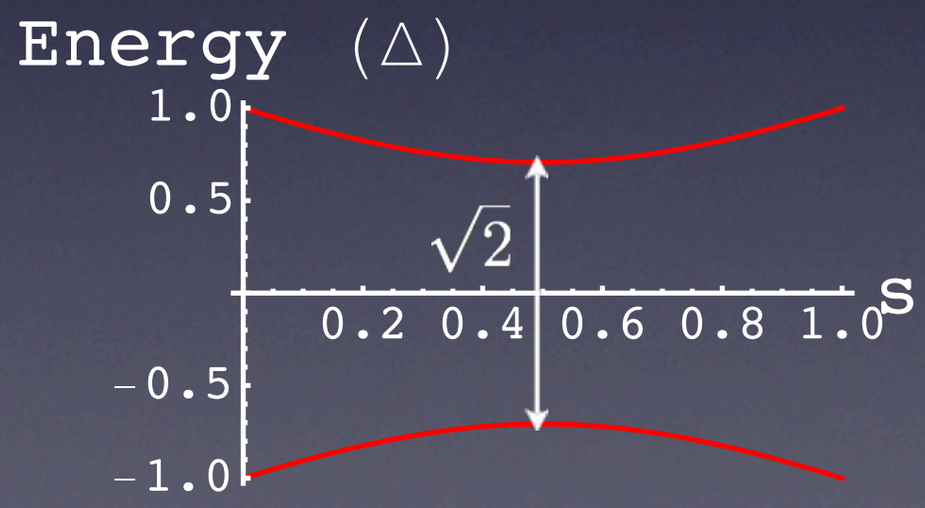
Initial Hamiltonian

$$H_i = -\Delta(\bar{Z}_1 + \bar{X}_2)$$

Final Hamiltonian

$$H_f = -\Delta(\bar{X}_1 + \bar{Z}_2)$$

$$H(s) = (1-s)H_i + sH_f$$



Adiabatic Teleportation

$$\begin{aligned} \bar{X}_1 &= X_1X_2 & \bar{Z}_1 &= Z_2Z_3 \\ \bar{X}_2 &= X_2X_3 & \bar{Z}_2 &= Z_1Z_2 \\ \bar{X}_3 &= X_1X_2X_3 & \bar{Z}_3 &= Z_1Z_2Z_3 \end{aligned}$$

Initial Hamiltonian

$$H_i = -\Delta(\bar{Z}_1 + \bar{X}_2)$$

Info in first qubit:

$$X_1 = \bar{X}_2\bar{X}_3$$
$$Z_1 = \bar{Z}_1\bar{Z}_3$$

Final Hamiltonian

$$H_f = -\Delta(\bar{X}_1 + \bar{Z}_2)$$

Info in third qubit:

$$X_3 = \bar{X}_1\bar{X}_3$$
$$Z_3 = \bar{Z}_2\bar{Z}_3$$

$$|\psi\rangle \otimes |\Phi_+\rangle \quad \rightarrow \quad |\Phi_+\rangle \otimes |\psi\rangle$$

Adiabatic Gate Teleportation

[Bacon and Flammia, Phys. Rev. Lett. 103, 120504 (2009)]

$$H_i = -U_3\Delta(X_2X_3 + Z_2Z_3)U_3^\dagger \qquad H_f = -\Delta(X_1X_2 + Z_1Z_2)$$

Note: spectrum unchanged by unitary

$$|\psi\rangle \otimes |\Phi_+\rangle \longrightarrow |\Phi_+\rangle \otimes U_3|\psi\rangle$$

Example (Hadamard gate): $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$H_i = -\Delta(X_2Z_3 + Z_2X_3) \qquad H_f = -\Delta(X_1X_2 + Z_1Z_2)$$

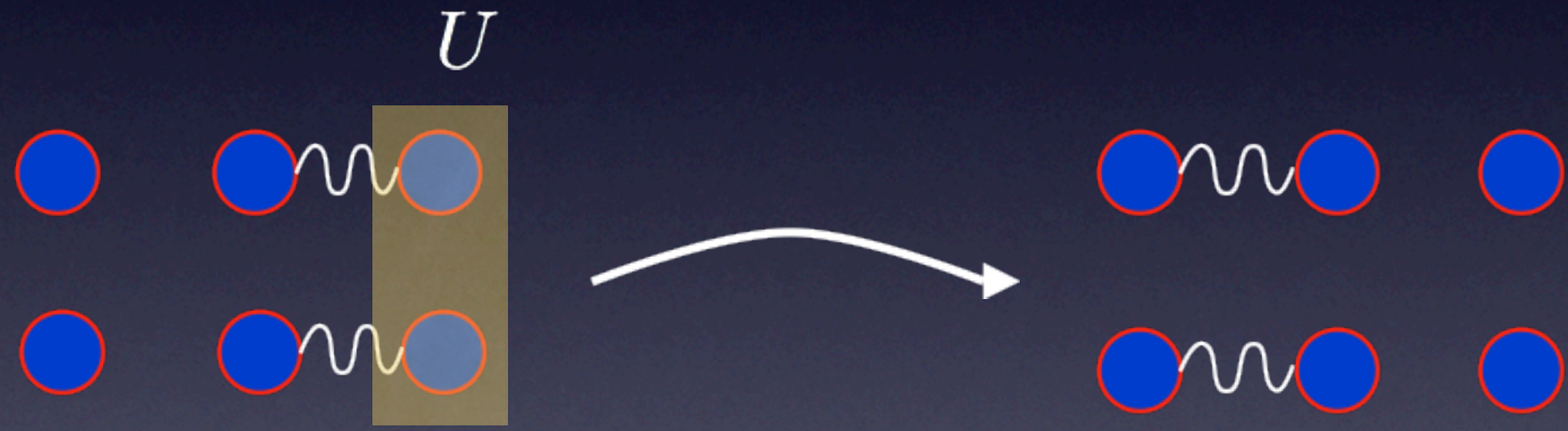
$$|\psi\rangle \otimes |\Phi_+\rangle \longrightarrow |\Phi_+\rangle \otimes H|\psi\rangle$$

Let's Build a Computer

One qubit gates



Two qubit gates



Problem: initial Hamiltonian requires 3 qubit interactions

Example



$$H_i = -\Delta(X_2X_3Z_6 + Z_2Z_3 + Z_5Z_6 + Z_3X_5X_6)$$

To get 2 qubit interactions: perturbation theory gadgets

Gadgets



Replace 3,6 by encoded qubits: $|0_L\rangle = |00\rangle, |1_L\rangle = |11\rangle$

$$
\begin{aligned}
H_i &= -\lambda(X_2X_{3a} + Z_2Z_{3a}) - \lambda(X_5X_{6a} + Z_5Z_{6a}) \\
&\quad -\omega Z_{3a}Z_{3b} - \lambda(X_{3b}Z_{6b} + Z_{3b}X_{6b}) \qquad \lambda \ll \omega \\
H_f &= -\lambda(X_1X_2 + Z_1Z_2) - \lambda(X_4X_5 + Z_4Z_5) \\
&\quad -\omega Z_{3a}Z_{3b}
\end{aligned}
$$

[S. D. Bartlett and T. Rudolph, Phys. Rev. A 74, 040302 (2006)]

Gadgets



$$
\begin{aligned}
H_i &= -\lambda(X_2X_{3a} + Z_2Z_{3a}) - \lambda(X_5X_{6a} + Z_5Z_{6a}) \\
&\quad -\omega Z_{3a}Z_{3b} - \lambda(X_{3b}Z_{6b} + Z_{3b}X_{6b}) \\
H_f &= -\lambda(X_1X_2 + Z_1Z_2) - \lambda(X_4X_5 + Z_4Z_5) \\
&\quad -\omega Z_{3a}Z_{3b}
\end{aligned}
$$

Minimum gap: $O\left(\frac{\lambda^2}{\omega}\right)$

Initial ground state fidelity: $1 - O\left(\frac{\lambda^2}{\omega^2}\right)$

Variations on a Theme

Exchange interactions fix this:

$$H = -\Delta(X_1X_2 + Y_1Y_2 + Z_1Z_2)$$

Adiabatically preparing rotated Hamiltonians:

$$H_i = -\Delta(X_1X_2 + Z_1Z_2) \qquad H_f = -\Delta U_2(X_1X_2 + Z_1Z_2)U_2^\dagger$$

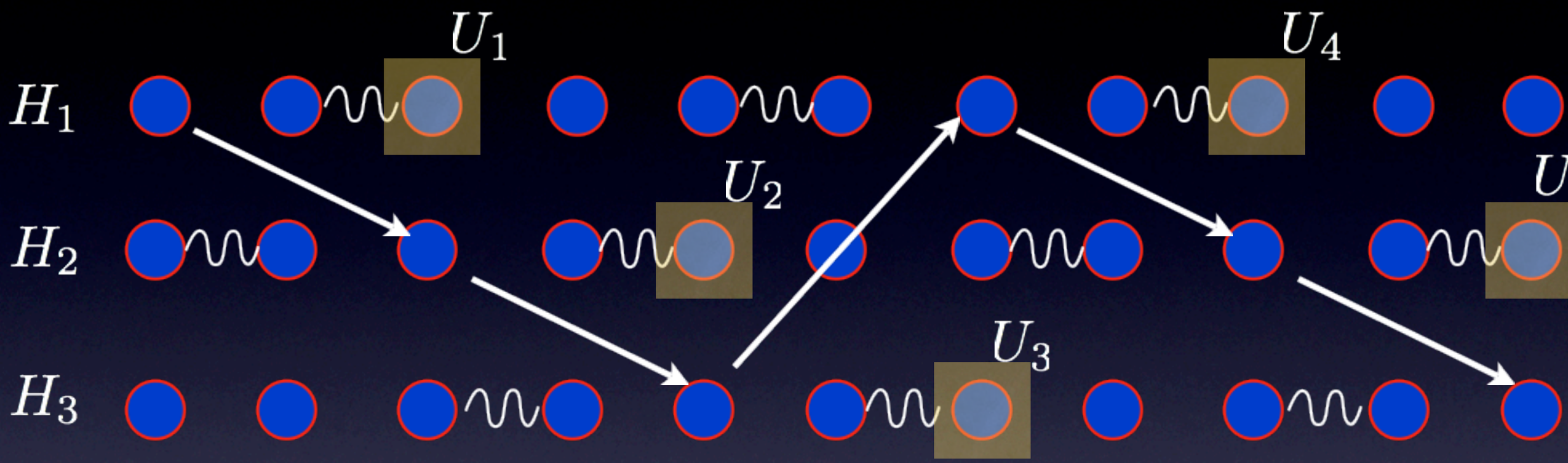
Not all U's allow a gap

$$A = \frac{1}{2}\begin{bmatrix} 1+i\sqrt{2} & 1 \\ 1 & -1+1\sqrt{2} \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{bmatrix}$$

gap: $\sqrt{2}\Delta$ gap: $\sqrt{2-\sqrt{2}}\Delta$

Architectures





Sequence of Hamiltonians: $H_1, H_2, H_3, H_1, H_2, H_3, \ldots$

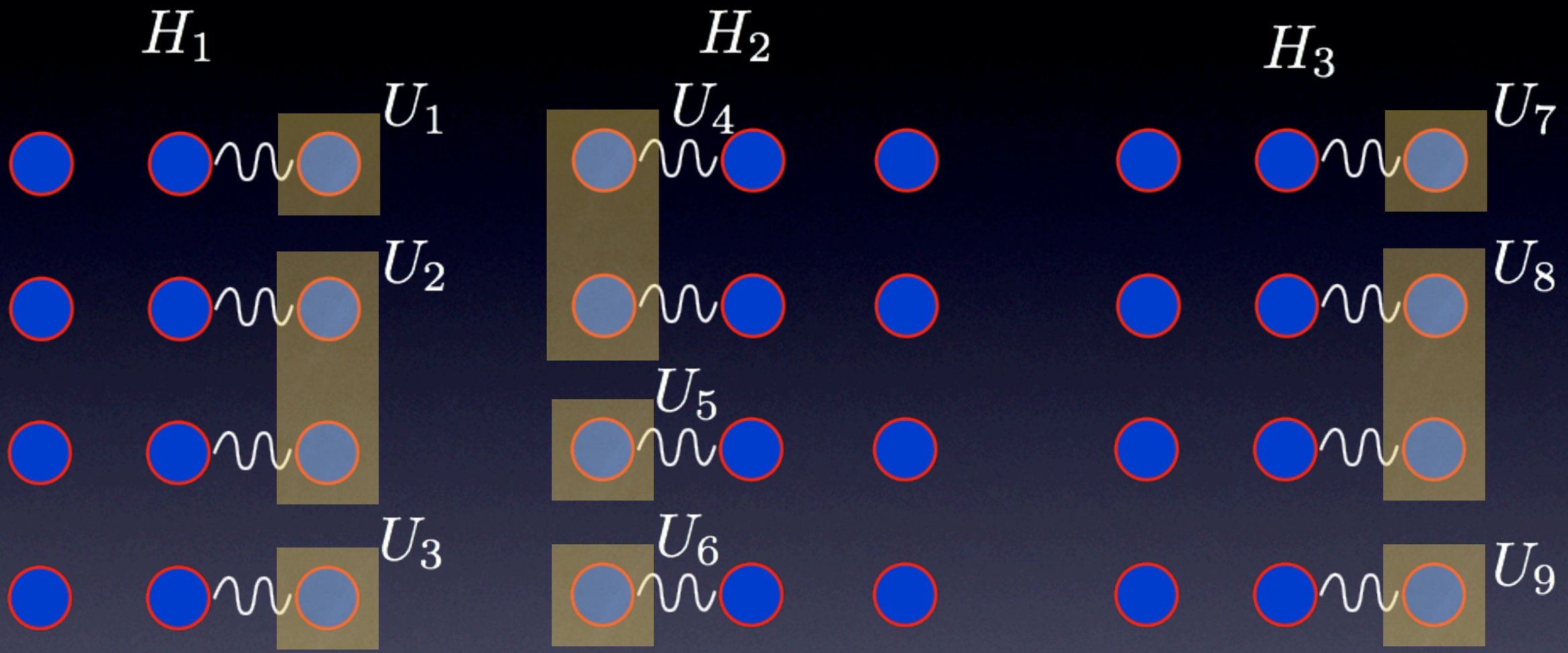
Sequence of single qubit gates applied: $U_1, U_2, U_3, U_4, U_5, \ldots$

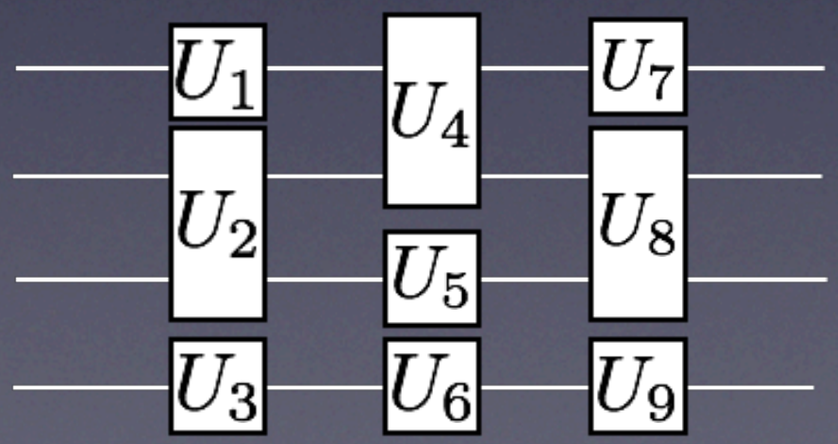
Generalizing to more than 1 qubit:

Universal QC by cycling between only 3 Hamiltonians

Architectures







n qubit circuit

4n or 5n qubits

Adiabatic Gate Teleportation

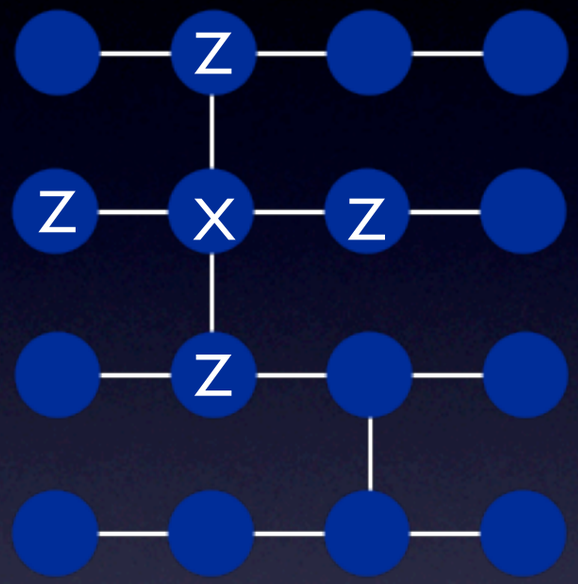
+ Simple adiabatic gates
+ Fault-tolerant (port proof from gate model)
- Gadget construction slows down 2-qubit gates
- Still requires state preparation and final measurement

Anything You Can Do I Can Do Adiabatically?

- Adiabatic Gate Teleportation takes a conditional measurement based protocol and performs it adiabatically

- Can other such protocols be similarly converted (like the one-way model?)

Cluster State Hamiltonian



$$S_v = X_v \prod_{w \text{ adjacent to } v} Z_w$$

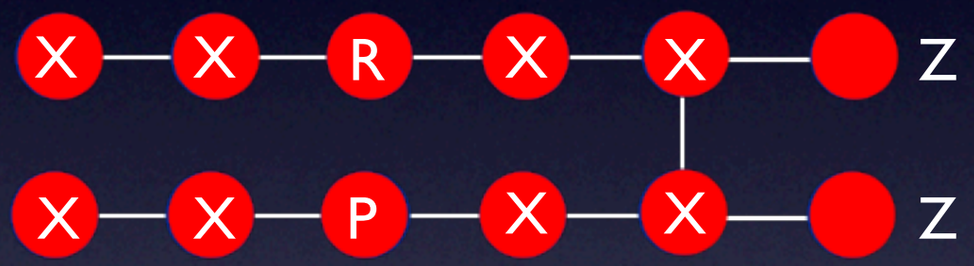
Cluster state is ground state of $H_C = -\Delta \sum_v S_v$

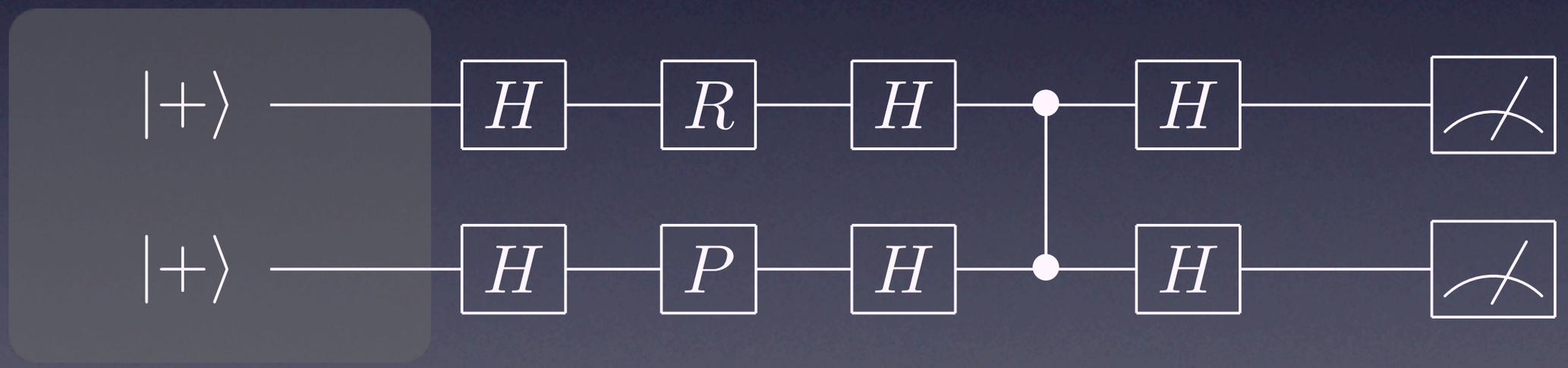
Possible to use gadgets to make only 2-qubit interactions:

[S. D. Bartlett and T. Rudolph, Phys. Rev. A 74, 040302 (2006)]

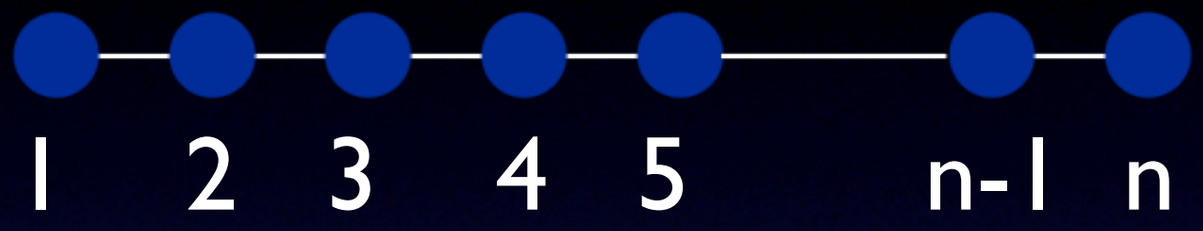
Adiabatic Cluster State QC

By turning on single qubit Hamiltonians while turning off parts of the cluster state Hamiltonian, we can enact a quantum circuit:





One Qubit



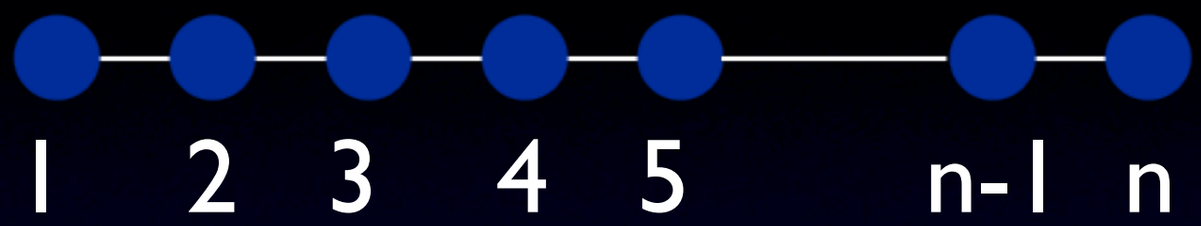
Modified cluster Hamiltonian:

$$H_C = -\Delta \sum_{i=1}^{n-1} S_i \qquad S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 1 \leq i \leq n-2$$
$$S_{n-1} = [Z]_{n-1}[X]_n$$

Ground state is 2-fold degenerate, can define logical operators for this qubit:

$$\bar{X} = [X]_1[Z]_2$$
$$\bar{Z} = [Z]_1$$

One Qubit



Modified cluster Hamiltonian:

$$H_C = -\Delta \sum_{i=1}^{n-1} S_i \qquad S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 1 \le i \le n-2$$

$$S_{n-1} = [Z]_{n-1}[X]_n$$

Adiabatically turn on field along X direction:

$$H_C + \Delta S_1 - \Delta [X]_1$$

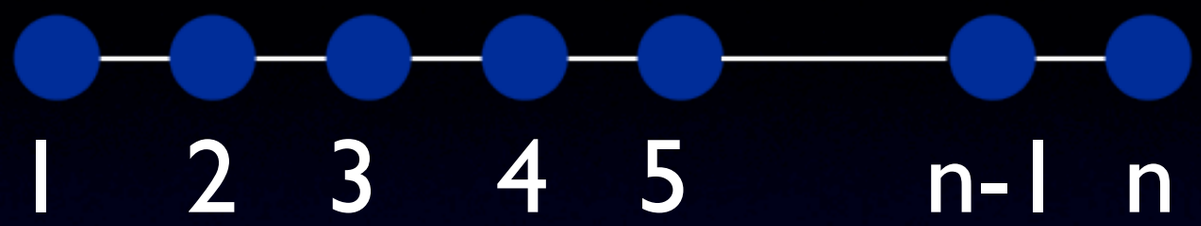
anticommute

$$\bar{X} = [X]_1[Z]_2 \qquad [X]_1 \text{ commute}$$

$$\bar{Z} = [Z]_1 \qquad \bar{Z} \cong [Z]_1 S_1 = [X]_2[Z]_3$$

anticommute

One Qubit



Modified cluster Hamiltonian:

$$H_C = -\Delta \sum_{i=1}^{n-1} S_i \qquad S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 1 \leq i \leq n-2$$

$$S_{n-1} = [Z]_{n-1}[X]_n$$

Adiabatically turn on field along X direction:

$$H_C + \Delta S_1 - \Delta [X]_1$$

Net result:

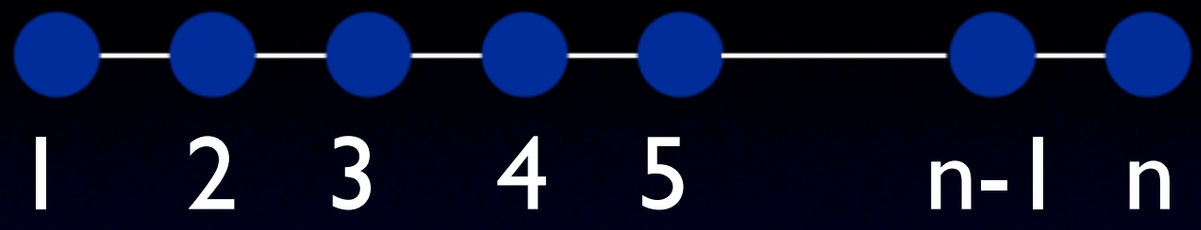
$$\bar{X} = [X]_1[Z]_2 \qquad \bar{Z} = [Z]_1$$



$$\bar{X} = [Z]_2 \qquad \bar{Z} = [X]_2[Z]_3$$



One Qubit Rotations



Modified cluster Hamiltonian:

$$H_C = -\Delta \sum_{i=1}^{n-1} S_i \qquad S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 1 \le i \le n-2$$

$$S_{n-1} = [Z]_{n-1}[X]_n$$

Adiabatically turn on field along rotated direction:

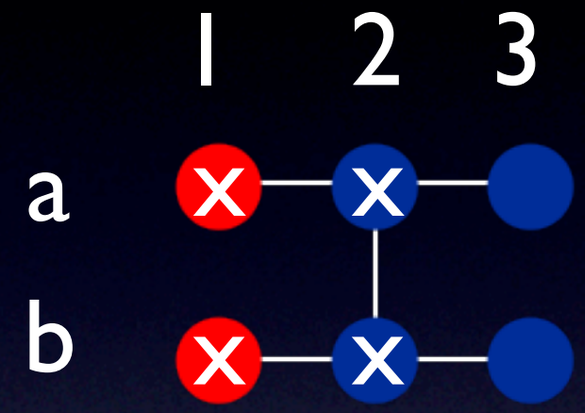
$$H_C + \Delta S_1 - \Delta(\cos\theta[X]_1 + \sin\theta[Y]_1)$$

Net result:



$$U(\theta) = \exp(i\theta/2Z)$$

Two Qubit Gate



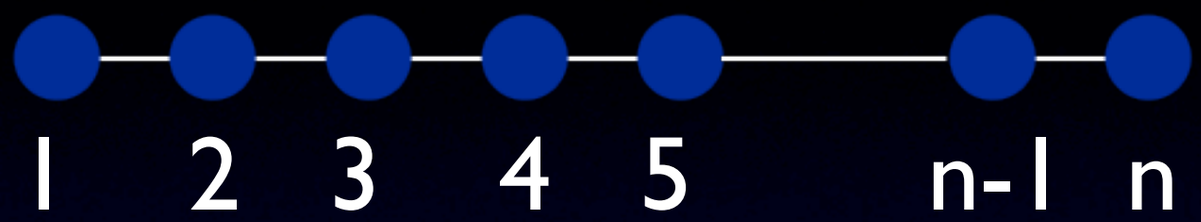
$$Z_a = [Z]_{1,a} \qquad Z_a \cong [X]_{2,a}[Z]_{3,a}[X]_{3,b} \qquad \bar{Z}_a \cong [Z]_{3,a}[X]_{3,b}$$

$$\bar{X}_a = [X]_{1,a}[Z]_{2,a} \qquad \bar{X}_a \cong [X]_{1,a}[X]_{3,a} \qquad \bar{X}_a \cong [X]_{3,a}$$

$$\bar{Z}_b = [Z]_{1,b} \qquad \bar{Z}_b \cong [X]_{2,b}[Z]_{3,b}[X]_{3,a} \qquad \bar{X}_b \cong [X]_{3,b}$$

$$\bar{X}_b = [X]_{1,b}[Z]_{2,b} \qquad \bar{X}_b \cong [X]_{1,b}[X]_{3,b} \qquad \bar{Z}_b \cong [Z]_{3,b}[X]_{3,x}$$

State Preparation



Cluster Hamiltonian:

$$H_C = -\Delta \sum_{i=1}^{n-1} S_i \qquad S_0 = [X]_1[Z]_2$$

$$S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 1 \leq i \leq n-2$$

$$S_{n-1} = [Z]_{n-1}[X]_n$$

Turn on two magnetic fields

$$H_C - \Delta([X]_1 + [X]_2 - S_0 - S_1 - S_3)$$

Not degenerate to degenerate!

But there is a conserved quantity: $S_1 S_3 \quad [X]_3[Z]_4$

Start in +1 e.v. of conserved quantity, so end there

So end in +1 e.v. of $[X]_3[Z]_4$

Adiabatic Cluster State QC

+ Simple adiabatic gates
+ Requires only final measurement (like topological quantum computing)
+ Fault-tolerant to model where cluster state is thermalized
- Gadget construction slows down 2-qubit gates

Models of QC



Anything you can do,
I can do adiabatically
I can do anything
slower than you!

But slower = speed set by energy gap (so not so slow)
And adiabatic = robust to timing errors

Given simple two-qubit adiabatic evolutions, plus final readout, we can build a quantum computer.

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